

Drew Villani

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EDUCATION

Bachelor of Science, Mechanical Engineering

University at Buffalo, The State University of New York

May 2028

GPA: 3.4/4.0

TECHNICAL SKILLS

CAD: Fusion 360, Autodesk Inventor, SolidWorks, Revit, Parametric Modeling, Generative Design, GD&T
Analysis: MATLAB, FEA, Nonlinear FEA, buckling, fatigue stress
Manufacturing: Machining, FDM, Arduino, DFMA, DFAM, material selection

ENGINEERING WORK EXPERIENCE

Product Engineering Intern, On Top Safety Co., Cheektowaga, NY

Aug 2025 – May 2026

- Developed two patented construction safety device concepts from initial requirements through CAD, analysis, prototyping using SolidWorks, MATLAB, nonlinear FEA, physical testing
- Reduced predicted fall-arrest force by 33% through parametric evaluation of spring rate, preload, stroke length, housing geometry, and energy-absorption architecture
- Built MATLAB tools to screen 20+ spring configurations against force, travel, packaging, design feasibility targets for mechanical energy-absorption system.

Project Lead, UB ASME SDC (*American Society of Mechanical Engineers*)

May 2026 – Present

IAM3D System Lead, UB ASME IAM3D

Aug 2025 – May 2026

- Led chassis development for a competition FPV payload rover, coordinating subsystem interfaces, design reviews, integration decisions across a 10+ member team
- Authored 160-page technical report covering design rationale, FEA assumptions, vibration considerations, electrical systems, failure risks
- Designed modular 3D-printed chassis using generative design for primary structural members, with crosslink-based support bars to manage load distribution and torsional stiffness
- Evaluated chassis members, mounting features, and crossmembers using FEA, hand calculations, and load-path reasoning prior to final integration
- Developed custom fasteners and snap-fit interfaces to eliminate hardware dependencies, reduce assembly time, and meet printability and serviceability constraints

ENGINEERING PROJECTS

Fall-Arrest Spring Housing Prototype

Fall 2025

- Designed a multi-part spring housing for a construction fall-arrest system using SolidWorks, MATLAB energy modeling, nonlinear FEA, rapid prototyping
- Evaluated spring stiffness, stroke length, material options, and housing geometry to minimize arrest force while maintaining practical packaging and manufacturability
- Performed **nonlinear FEA, buckling analysis, fatigue stress evaluation** on spring and housing geometry to characterize load-deflection behavior and assess failure risk under cyclic loading

OTHER EXPERIENCE

Mechanical & Electrical Diagnosis and Repair (Self-Employed)

Jun 2023 – Aug 2025

- Bought, diagnosed, repaired, resold 14 vehicles over 2 years, generating 5-figure profit through cost-controlled repair decisions
- Diagnosed engine, suspension, brake, steering, and electrical faults using OBD-II data, service information, inspection, and hands-on testing